Course Notes

Lie Theory

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Chapter 1

Lecture 1

1.1 Introduction

Lie algebras were originally introduced by Sophus Lie in the 1870s to study the concept of continuous transformation groups. Lie's motivation was to develop a theory for differential equations analogous to Galois theory for algebraic equations. He discovered that the infinitesimal transformations of a continuous group form a linear space equipped with a bracket operation, now known as a Lie algebra. This linearization allows complex global geometric problems to be translated into algebraic problems, making them more tractable.

Let F be a field (mostly $\operatorname{char} F = 0$, even $F = \mathbb{C}$).

Definition 1.1.1. *A linear algebra A is a vector space over F, with a binary bilinear operation:*

- *binary operation: $A \times A \to A$, $(a, b) \mapsto ab$.*
- *bilinear: this operation is linear in each variable. In orther word, $\forall \alpha \in F$ and $a, a', a'', b, b', b'' \in A$, we have:*

$$\begin{aligned}(\alpha a)b &= a(\alpha b) = \alpha(ab)\\ (a' + a'')b &= a'b + a''b\\ a(b' + b'') &= ab' + ab''\end{aligned}$$

The algebra A is associative if $\forall a, b, c \in A$, $(ab)c = a(bc)$.

Example 1.1.2. *$M_n(F)$, the algebra of $n \times n$ matrices over F.*

Example 1.1.3. *Let V be a vector space, $\operatorname{End}_F(V) = \{$linear transformations $\varphi : V \to V\}$ is an associative algebra with the composition operation$(\phi\psi)(v) = \phi(\psi(v))$. If $\dim V < \infty$, when we fix a basis, then we fet a matrix representation of φ in our fix basis.*

The most imprtant thing everyone shoul know is Wedderburn's theorem.

Theorem 1.1.4 (Wedderburn-Artin)**.** *Let A be a semisimple Artinian ring (with 1). Then A is a direct sum of matrix rings over division rings.*
Specifically, if ${}_AA \simeq S_1^{n_1} \oplus \cdots \oplus S_r^{n_r}$ where $S_1, \ldots, S_r$ are non-isomorphic simple modules occurring with multiplicities $n_1, \ldots, n_r$, then

$$A \simeq M_{n_1}(D_1) \oplus \cdots \oplus M_{n_r}(D_r)$$

where $D_i = \mathrm{End}_A(S_i)^{op}$. This is a ring isomorphism, and each $M_{n_i}(D_i)$ is a 2-sided ideal.

Proof. First, since A is a semisimple ring, its left regular module ${}_AA$ can be decomposed into a direct sum of simple left A-modules:

$${}_AA \simeq S_1^{n_1} \oplus \cdots \oplus S_r^{n_r}$$

where $S_1, \ldots, S_r$ are pairwise non-isomorphic simple modules.

By Schur's lemma, the endomorphism ring of a simple module is a division ring, so $D_i = \mathrm{End}_A(S_i)^{op}$ is a division ring.

Next, we compute the endomorphism ring of the left regular module $\mathrm{End}_A({}_AA)$. On one hand, for any ring A with 1, we have the ring isomorphism $\mathrm{End}_A({}_AA) \simeq A^{op}$. On the other hand, using our direct sum decomposition:

$$\mathrm{End}_A({}_AA) \simeq \mathrm{End}_A(S_1^{n_1}) \oplus \cdots \oplus \mathrm{End}_A(S_r^{n_r})$$

This decomposition holds because $\mathrm{Hom}_A(S_i^{n_i}, S_j^{n_j}) = 0$ whenever $i \neq j$.

For each component in the direct sum, we can rewrite the endomorphism ring using matrices over D_i:

$$\mathrm{End}_A(S_i^{n_i}) \simeq M_{n_i}(D_i)^{op} \simeq M_{n_i}(D_i^{op})$$

Substituting this back into our equation for A^{op}, we obtain:

$$A^{op} \simeq M_{n_1}(D_1^{op}) \oplus \cdots \oplus M_{n_r}(D_r^{op})$$

To recover the ring A, we take the opposite ring on both sides. Recall the property that $M_n(R)^{op} \simeq M_n(R^{op})$. Applying this to each block yields:

$$A \simeq (A^{op})^{op} \simeq \left(\bigoplus_{i=1}^{r} M_{n_i}(D_i^{op}) \right)^{op} \simeq \bigoplus_{i=1}^{r} M_{n_i}(D_i)$$

Thus, the isomorphism is established. □

Remark 1.1.5. *If the base field F is algebraically closed, every finite-dimensional division* *algebra over F is equal to F. Hence the theorem simplifies to*

$$A \cong \prod_{i=1}^{r} M_{n_i}(F).$$

Example 1.1.6. *Let G be a finite group and let $\mathbb{C}[G]$ be its group algebra. By Maschke's theorem $\mathbb{C}[G]$ is semisimple. Therefore*

$$\mathbb{C}[G] \cong \prod_{i=1}^{r} M_{n_i}(\mathbb{C}),$$

where n_i are the dimensions of the irreducible complex representations of G.

Definition 1.1.7. *Suppose F is algebraically closed. An algebra A is simple if*

1. $\nexists\ I \lhd A$ *and* $I \neq 0$.
2. $A \cdot A \neq (0)$.

Corollary 1.1.8. *Any simple finite dimensional associative algebra is isomorphic to $M_n(F)$ for some n.*

Definition 1.1.9 (Jacobson Radical)**.** *Let A be an associative algebra with identity over a field F. The* Jacobson radical *of A, denoted by* $\mathrm{Rad}(A)$ *or $J(A)$, is defined as the intersection of all maximal left ideals of A:*

$$\mathrm{Rad}(A) = \bigcap_{M \text{ maximal left ideal of } A} M.$$

Proposition 1.1.10. *Let A be an associative algebra with identity. Then*

$$\mathrm{Rad}(A) = \bigcap_{S \text{ simple left } A\text{-module}} \mathrm{Ann}_A(S),$$

where

$$\mathrm{Ann}_A(S) = \{a \in A \mid aS = 0\}$$

is the annihilator of S in A.

Proposition 1.1.11 (Equivalent Characterization)**.** *Let A be an associative algebra with identity. Then an element $a \in A$ lies in* $\mathrm{Rad}(A)$ *if and only if for every $x \in A$ the element*

$$1 - ax$$

is invertible in A.

Proposition 1.1.12. *The Jacobson radical* $\mathrm{Rad}(A)$ *is a two-sided ideal of* A*.*

Definition 1.1.13 (Semisimple Algebra)**.** *An associative algebra* A *is called* semisimple *if*

$$\mathrm{Rad}(A) = 0.$$

1.2 Where do Lie algebras come from?

Definition 1.2.1. *A Lie algebra is a vector space* L *over a field* F*, with a binary operation* $[a, b] \in L$*, such that*

1. $[a, b]$ *is bilinear.*
2. $[a, b] = -[b, a]$*.*
3. $[a, [b, c]] + [c, [a, b]] + [b, [c, a]] = 0$*, this is called Jacobi identity.*

1.2.1 They come from associative (central simple) algebras

Let A be an associative algebra. $\forall a, b \in A$, define $[a, b] = ab - ba$. Check that $(A, [\ ,\ ])$ is a Lie algebra:

Proof. Bilinearity and skew-symmetry are clear. For Jacobi identity:

$$\begin{aligned}
&[a, [b, c]] + [b, [c, a]] + [c, [a, b]] \\
&\qquad = a(bc - cb) - (bc - cb)a + b(ca - ac) - (ca - ac)b + c(ab - ba) - (ab - ba)c \\
&\qquad = abc - acb - bca + cba + bca - bac - cab + acb + cab - cba - abc + bac \\
&\qquad = 0.
\end{aligned}$$

□

We denote this new algebra $(A, [\ ,\ ])$ by $A^{(-)}$.
Suppose that $L \subseteq A$ and $[L, L] \subseteq L$. Then $L \leq A^{(-)}$。

Example 1.2.2. $M_n(F)^{(-)} = \mathfrak{gl}(n)$*. We call it the general linear Lie algebra.*
$[L, L] = \{\sum_i [a_i, b_i]\} \trianglelefteq L$*. Clearly* $[\mathfrak{gl}(n), \mathfrak{gl}(n)] \neq \mathfrak{gl}(n)$ *and for* $n \geq 2$*,* $[\mathfrak{gl}(n), \mathfrak{gl}(n)] \neq 0$*.*

Example 1.2.3. *Focus on* $[\mathfrak{gl}(n), \mathfrak{gl}(n)]$*, all matrices are of trace 0. This is called the special linear Lie algebra, denoted by* $\mathfrak{sl}(n)$*.*

Definition 1.2.4. Z *is the center of Lie algrbra* L *if* $Z = \{z \in L | [z, L] = 0\}$*.*

1.2.2 They come from involutions of central simple algebras

We can also find Lie algebras from the involution of a central simple algebra. Let A be central simple algebra and $* : A \to A$ be a linear transformation. $*$ is called a an involution if $(a^*)^* = a$ and $(ab)^* = b^* a^*$, $\forall a, b \in A$.

Proposition 1.2.5. *Let A be a central simple algebra over a field F equipped with an involution $\sigma : A \to A$. Then the set*

$$L = \{x \in A \mid \sigma(x) = -x\}$$

forms a Lie algebra under the commutator bracket

$$[x, y] = xy - yx.$$

Proof. First observe that any associative algebra A becomes a Lie algebra with the commutator bracket

$$[x, y] = xy - yx.$$

Let

$$L = \{x \in A \mid \sigma(x) = -x\}.$$

We show that L is closed under the Lie bracket.

Take $x, y \in L$. Then

$$\sigma(x) = -x, \qquad \sigma(y) = -y.$$

Since σ is an involution, it satisfies

$$\sigma(xy) = \sigma(y)\sigma(x).$$

Hence

$$\sigma([x, y]) = \sigma(xy - yx) = \sigma(y)\sigma(x) - \sigma(x)\sigma(y).$$

Substituting $\sigma(x) = -x$ and $\sigma(y) = -y$, we obtain

$$\sigma([x, y]) = (-y)(-x) - (-x)(-y) = yx - xy = -[x, y].$$

Thus $[x, y] \in L$. Therefore L is closed under the commutator bracket.

Since the commutator bracket in any associative algebra satisfies bilinearity, antisymmetry, and the Jacobi identity, it follows that L is a Lie algebra. $\square$

Example 1.2.6. *The Quatanion algebra is an associative algebra with the involution $a \mapsto a^*$. It is 4 dimensional central simple algebra with standard basis $1, i, j, k$, where* *the relations are* $i^2 = j^2 = k^2 = -1, ij = k, jk = i, ki = j$ *as the following graph:*

$$\begin{tikzcd} i \arrow[r] & j \arrow[d] \\ & k \arrow[ul] \end{tikzcd}$$

We use $\mathbb{H}$ *to denote the Quatanion algebra when we choose* $\mathbb{R}$ *as the base field. i.e.* $\mathbb{H} = \{a_0 1 + a_1 i + a_2 j + a_3 k | a_0, a_1, a_2, a_3 \in \mathbb{R}\}$. $\forall x \in \mathbb{H}$ *where* $x = a_0 1 + a_1 i + a_2 j + a_3 k$, $*$ *is defined as* $x^* = a_0 1 - a_1 i - a_2 j - a_3 k$ *and* $x^{-1} = \frac{1}{a_0^2 + a_1^2 + a_2^2 + a_3^2}(a_0 1 - a_1 i - a_2 j - a_3 k)$. *Hence* $\mathbb{H}$ *is a division algebra.*

If the base field is algebraically closed, $F = \overline{F}$. *We can denote the Quatanion algebra by* 2×2 *matrices*

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \xi \end{pmatrix}$$

Where $\alpha, \beta, \gamma, \xi \in F$. *In this case it is no longer a division algebra* ($a_0^2 + a_1^2 + a_2^2 + a_3^2$ *may equal to 0), but just* $M_2(F)$. *And in this case the involution* $*$ *is defined as:*

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \xi \end{pmatrix} \mapsto \begin{pmatrix} \xi & -\beta \\ -\gamma & \alpha \end{pmatrix}$$

Which is the Simplectic involution.

Example 1.2.7.

Chapter 2

Lecture 2

Recall last time:

1. Matrix Lie algebras, e.g. $\mathfrak{sl}(n)$. Consider involution $* : M_n(F) \to M_n(F)$ skew-symmetric matrices.

2. Derivation, $\mathrm{Der}(A)$. e.g. $A = F[t_1, \dots, t_n]$, $\mathrm{d} = \sum_{min}^{max} f_i()\partial$

A non-associative algebra, $\mathrm{d} : x \mapsto [a, x]$ is a derivation.

Let L a Lie algebra, inner derivation

Q: d a derivation and exp(d). It need convergence.

Example 2.0.1. $\mathrm{d} = t_1 t_2 \frac{\partial}{\partial t_1}$.

3. Brackets:

2.1 Engel Theorem

L^n: span of products of elements of length n. Recall $\mathrm{ad}([a, b]) = [\mathrm{ad}(a), \mathrm{ad}(b)]$. $L \to \mathrm{End}_F(L)^{(-)}, a \mapsto \mathrm{ad}(a)$.

$[I, J]$ is again an ideal.

Decending chain:

$$L = L^1 \supseteq L^2 \supseteq \dots$$

L is nilpotent if....

eg: strict upper matrix

Another decending chain: $L^{[1]} = [L, L]$, $L^{[n+1]} = [L^{[n]}, L^{[n]}]$

L is solvable if .....

e.g. 2 dimentional sovable non nilpotent Lie alg.

A associa. $L \subseteq A^{(-)}$. Suppose $A =< L >$ as an associa alge. Then A is an enveloping algebra of L.

Theorem 2.1.1. *Let A be a finite dimensional associative algebra, $L \subseteq A^{(-)}$, $A =< L >$. $\exists n \geq 1$, $\forall a \in L$, $a^n = 0$. Then A is nilpotent.*

We say that a subset $S \subseteq L$ is a Lie subset if $\forall a, b \in S$, $[a, b] \in S$.

Theorem 2.1.2. *Let A be a finite dimensional associative algebra, $L \subseteq A^{(-)}$, $A =< L >$. Let S be a Lie subset of L. $L = \mathrm{span}(S)$, $\forall a \in S$, $a^n = 0$.*

Choose a maximal Lie subset $S_1 \subset S$ such that the associative $< S_1 >$ is nilpotent. So $\exists n$ such that $S_1 \dots S_1 = (0)$ (n times) (existance?)

claim $\mathrm{ad}(S_1) \dots \mathrm{ad}(S_n) = (0)$.